

Weekly Notice 7

Exercises, week 7:

1. Exercise 62.
2. Exercise 49. In part (2) only prove that B_1 and B_2 are independent under $\mathbb{P}_i$, and that $\mathbb{P}_i(B_1) = \mathbb{P}_i(B_2)$.
3. Exercise 72.
4. Exercise 73.
5. The exercise below.
6. Discuss the content of Lemma 6.16, Definition 6.25, and Proposition 6.26. Also relate to Exercise 7(3).
(As background, I can mention the detailed balance equation also plays an important role in the Metropolis-Hastings algorithm, widely used in Markov chain Monte Carlo (MCMC)).
7. Exercise 77.

Exercise: Two-state transition semigroup and jump rates.

Let $S = \{1, 2\}$. Fix $\alpha, \beta \geq 0$ and set $\lambda = \alpha + \beta$ (assume $\lambda > 0$ unless stated otherwise). For $t \geq 0$ define the transition probabilities $p_t = \{p_t(x, y)\}_{x, y \in S}$ as

$$\begin{aligned} p_t(1, 1) &= \frac{\beta}{\lambda} + \frac{\alpha}{\lambda} e^{-\lambda t}, & p_t(1, 2) &= \frac{\alpha}{\lambda} (1 - e^{-\lambda t}), \\ p_t(2, 1) &= \frac{\beta}{\lambda} (1 - e^{-\lambda t}), & p_t(2, 2) &= \frac{\alpha}{\lambda} + \frac{\beta}{\lambda} e^{-\lambda t}. \end{aligned}$$

1. Show that $(p_t)_{t \geq 0}$ is a transition semigroup.
2. Often, one speaks of the *rate of a transition from state 1 to state 2* (or vice versa). How would you define this rate? Determine the rate.
3. Consider a Markov process with the transition semigroup above. Assume the process starts in state 1. Intuitively, what is the distribution of how long the process stays in state 1 before it jumps to state 2?
4. Do you dare describing how the process $(Y_t)_{t \geq 0}$ evolves over time?